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Poor sleep quality among bedtime smartphone user medical students in Ethiopia, 2024

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ABSTRACT

Background: Smartphone use has become widespread and continues to grow rapidly. Medical students, already highly susceptible to sleep deprivation, may experience exacerbated issues due to smartphone usage, particularly at bedtime. Therefore, understanding the potential negative impacts of this behavior is crucial. This study aims to assess the prevalence and risk factors of sleep quality among medical students bed time smart phone users in Ethiopia.

Subject and methods: An institutional-based cross-sectional study design was employed, involving 354 undergraduate medical students selected through simple random sampling from Debre Marko's University, the University of Gondar, and Debre Berhan University. Data were collected using the Pittsburgh Sleep Quality Index and structured interviews. Variables with a p-value of ≤0.2 in bivariable regression were included in multivariable logistic regression. Crude odds ratios and adjusted odds ratios were calculated, while chi-square tests were used to assess assumptions. In multivariable regression, variables with a p-value of ≤0.05 were deemed significant predictors at the 95 % confidence interval.

Result: The results showed that among bedtime smartphone users, 67.52 % had a poor sleep quality index greater than 5. Poor sleep quality was significantly linked to factors such as sex, regular coffee consumption, the purpose of smartphone use, phone position during use, the duration of smartphone use, and a history of disease. Social media usage was the most common activity, accounting for 41.60 % of smartphone use, followed by watching videos (21.65 %).

Conclusion: This study highlights the detrimental effects of bedtime smartphone use on sleep quality, which subsequently impacts mental. Given these findings, it is strongly recommended that medical students reduce their smartphone use before bedtime to improve their sleep quality.

1. Introduction

Poor sleep quality is defined by a Pittsburgh Sleep Quality Index (PSQI) score greater than five [1]. Sleep is essential for nearly all living organisms, as it supports critical processes like memory consolidation, body healing, and metabolic regulation [2]. Medicine is one of the most stressful fields of education due to its rigorous demands, which may contribute to the high prevalence of poor sleep quality among medical students. This issue is notably more severe in this group, with poor sleep

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Abbreviations

AOR	Adjusted Odds Ratio
CI	Confidence Interval
COR	Crude Odds Ratio
PSQI	Poor Sleep Quality Index

quality being twice as common among medical students compared to the general population [3]. For medical students, whose academic success depends heavily on optimal cognitive function and physical well-being, good sleep quality is particularly vital. However, late-night smartphone use, often perceived as a stressor, can significantly impair sleep quality, leading to diminished cognitive function and, consequently, a decline in academic performance [4].

Various studies have demonstrated that individuals who use smartphones at bedtime have a significantly higher prevalence of increased sleep latency (the time it takes to fall asleep), difficulty maintaining sleep (sleep disturbances), and reduced sleep duration compared to those who do not use smartphones before bed. These findings underscore the detrimental effects of bedtime smartphone use on overall sleep quality [5]. Smartphones have revolutionized global lifestyles and become especially crucial for students, offering connectivity, information access, web browsing, entertainment, and communication [6]. However, nighttime smartphone use has been associated with several circadian sleep-wake rhythm disorders, including insomnia and excessive daytime tiredness. These disruptions can have a significant impact on overall health and well-being, particularly in younger populations [7].

Prolonged nighttime use of mobile phones can significantly disrupt circadian rhythms. The blue light emitted from smartphone screens strongly stimulates retinal ganglion cells containing melanopsin, leading to the suppression of melatonin secretion. This hormonal disruption negatively impacts sleep quality, making it more difficult for individuals to fall asleep and maintain restful sleep throughout the night [2].

Globally, the prevalence of poor sleep quality among university students is approximately 18.5 %, significantly higher than the 7.4 % observed in the general population [8]. A systematic review and meta-analysis of African students found that the overall pooled prevalence of internet addiction was 34.53 % [9]. Similarly, smartphone addiction was prevalent among undergraduate university students in Ethiopia, with a rate of 53.6 % [10]. Additionally, the pooled estimate of poor sleep quality in Ethiopian people was 53 % [11]. Various studies have identified risk factors for poor sleep quality, including age, sex, monthly income, cigarette smoking, chat chewing, chronic diseases, smartphone utilization, religion, years of education, and obesity [12, 13]. However, there is a lack of published data on the prevalence and risk factors of sleep quality specifically among medical university students in Ethiopia. Therefore, this study aims to investigate the prevalence and risk factors of sleep quality among medical students bedtime smartphone users in Ethiopia.

1.1. Methodology

1.1.1. Study design, area, and period

A multicenter, institutional-based cross-sectional study was conducted among undergraduate university students in the Amhara region of Northwest Ethiopia from February to April 2024. The Amhara region is one of Ethiopia's largest regions, located approximately 565 km from Addis Ababa, the capital city of Ethiopia. The region hosts seven universities offering medical education: Bahir Dar University, Debre Tabor University, Debre Berhan University, University of Gondar, Wollo University, Woldia University, and Debre Marko's University. Through a simple random sampling technique, Debre Marko's University, University of Gondar, and Debre Berhan University were selected for the study.

1.2. Population

1.2.1. Source population

The source population for this study comprised all undergraduate university students in Amhara region.

1.2.2. Study population

The study population included all undergraduate university students in selected universities.

1.2.3. Study units

The study units are each student in which the actual data was collected.

1.3. Eligibility criteria

1.3.1. Inclusion criteria

- Undergraduate medical students currently enrolled as regular students in a medical program.
- Students willing to provide informed consent to participate in the study.

1.3.2. Exclusion criteria

- Students with serious health conditions or chronic illnesses that could significantly impact sleep quality or the study's results.
- Students using medications that are known to affect sleep patterns significantly and may skew results.

1.4. Sampling techniques and sample size estimation

The sample size of the study was determined using a single population proportion formula by considering the following: a proportion of 64.2 % of previous similar study performed in Egypt, CI: 95 %, and margin of error (d) = 5 %.

$$ni = \frac{(Z\alpha/2)^2 * P(1-P)}{d^2} = \frac{(1.92)2 * 0.642(1-0.642)}{(0.05)^2} = 354$$

With a 5 % contingency rate, the final sample size was 371.

During the data collection period, a total of 371 medical students were recruited across three universities: Debre Marko's University (323 students), University of Gondar (846 students), and Debre Berhan University (348 students). Using proportionate random sampling, 79 students were selected from Debre Marko's University, 207 from the University of Gondar, and 85 from Debre Berhan University.

1.5. Variables

Dependent variable: Sleep quality.

Independent variables: Sociodemographic characteristics like age, sex, residence, academic year, religion and family income were assessed. Behavioral and medical elements such as duration of smartphone use at bed, duration of daily use, purpose of use, year of smartphone use, adjustment of screen light, caffeinated drinking utilization, cigarette smoking, alcohol drinking, regular physical exercise, chronic disease, depression anxiety and stress disorder (DASS) and body mass index.

1.6. Operational definition

Smartphone overuse-was defined as using the device for more than 2 hours at bedtime [14].

Normal sleep duration-were defined as adults having a minimum of 7–8 hours of sleep per night [15].

Sleep quality- Good sleep quality is defined as having a PSQI score of ≤5, while individuals with a PSQI score >5 are considered to have poor sleep quality [1].

Consuming caffeinated beverages: Participants were first asked if they consumed more than one drink per week each month during the current academic semester/quarter (no vs. yes). Caffeinated beverages included coffee and Coke and Pepsi [16].

Body mass index: according to WHO BMI is classified as underweight, normal weight, overweight, and obese when weight over height is < 18.5 kg/m^2, 18.5–24.9 kg/m^2, 25–29.9 kg/m^2, and ≥30 kg/m^2 [17].

Smoker: A smoker is defined as someone who has smoked cigarettes at least once within the last 30 days [18].

Chewing khat: A current chewer is a person who has a history of khat chewing within the past one month [18].

1.7. Data collection procedures and instruments

Structured interviewer-administered questionnaires were utilized to collect data from participants (Supplementary file 1). The questionnaire, originally crafted in English, underwent translation into the participant's local language (Amharic). To ensure accuracy, experts proficient in both languages then back translated the translated version into English. Three individuals with Bachelor of Science degrees in health science conducted the interviews under the supervision of a principal investigator.

1.8. Measurements of sleep quality

The students' sleep quality was assessed by using the Vietnamese PSQI (PSQI-V) [19]. The PSQI consists of nineteen items grouped into seven categories: sleep duration, sleep disturbance, sleep latency, daytime dysfunction, sleep efficiency, subjective sleep quality, and use of sleeping medication. It aims to assess students' sleep quality over the previous month. Each component is scored from 0 to 3, with a total score ranging from 0 to 21. For each component, a score of 0 indicates no sleep difficulty, 1 indicates mild difficulty, 2 indicate severe difficulty, and 3 indicate very severe difficulty in sleep. Higher scores indicate poorer sleep quality. A total score greater than 5 indicates poor sleep quality, while a score equal to or less than 5 indicates good sleep quality [20]. The overall internal reliability of Cronbach's alpha coefficient for the PSQI-V in this study was 0.75.

1.9. Data quality control

To guarantee the data quality, data collectors underwent three days of training. To cheek the validity of the questionnaire Cronbach's alpha was examined, and the result was 75 %. Additionally, 5 % of the total sample size of medical students at Ambo university participated in the questionnaire pretesting. The main investigator closely monitored the data collectors each day. Before entry, the data underwent a thorough check for both consistency and completeness.

1.10. Data processing and analysis

Once the data was collected, it was exported into STATA version 14.0 for analysis. Descriptive statistics, including frequency, mean, standard deviation and percentage, were utilized to provide a comprehensive summary of the participants. The normality of the data was examined using the Shapiro-Wilk test, revealing a p-value of 0.004. Furthermore, the goodness-of-fit for the model was assessed through the Hosmer-Lemeshow test, with a calculated value of 0.264. Variables with a p-value of ≤0.2 in bivariable regression were included in multivariable logistics regression. To assess the strength of the association between the independent variables and outcome variables, crude odds ratios (COR) and adjusted odds ratios (AOR) at a 95 % confidence interval (CI) were calculated. In the multivariable regression, variables with a p-value of ≤0.05 were regarded as significantly associated.

2. Results

2.1. Sociodemographic characteristics of study participants

In this study, a total of 351 participants were selected using a systematic random sampling approach, resulting in an impressive 94.61 % response rate. The participants' ages ranged from 18 to 27 years, with a mean age of 22.03 ± 0.13. In terms of religious affiliation, 78 % identified as Christians. Regarding educational attainment, 29.34 % were first-year students. Additionally, 48.72 % of participants indicated residing in rural areas, while 28.57 % identified as civil servants. The mean monthly income among respondents was 6163.82 ± 97.69 ETB (Table 1).

2.2. Clinical and behavioral characteristics of the study participants

Among the participants selected, all were smartphone users who used their phones while in bed during sleep. Of these, 67.52 % used their phones for more than 2 h at bedtime, and 64.67 % used them daily. From the total study participants, 50.71 % did not adjust the screen display light at bedtime, 47.29 % placed their phone on the bed during sleep, and 41.60 % used their phones for social media. Additionally, 48.72 % had been using digital devices for five to ten years. Concerning sleep duration, 54.42 % of participants slept for less than 8 h. Among the study participants, 32.48 % engaged in physical exercise. Furthermore, 41.60 % of the total participants were regularly consuming caffeinated beverages, 9.97 % were current tobacco smokers, and 35 % were alcohol drinkers. Notably, 28.77 % of the participants reported experiencing stress and anxiety disorders, while 27.64 % had chronic diseases. In terms of BMI, 76.35 % of respondents fell within the normal range (Table 2).

2.3. Prevalence of poor sleep quality among health science students

The results of this study revealed that 67.52 % (95 % CI: 62.42–72.24) of the participants had poor sleep quality (Fig. 1).

2.4. Factors associated with poor sleep quality

During the bivariable analysis, factors such as sex, physical exercise,

Table 1
Sociodemographic characteristics of study participants in Ethiopia, 2024.

Variables	Category	Full sample		Good sleep quality		Poor sleep quality	
		N	%	N	%	N	%
Age	18–27	351	100	114	32.48	237	67.52
Sex	Male	196	55.84	89	78.01	107	45.15
	Female	155	44.16	25	21.93	130	54.85
Religion	Christiane	273	78	75	65.79	198	83.54
	Muslim	78	22	39	34.21	39	16.46
Academic year	1st	103	29.34	44	38.60	59	24.89
	2nd	80	22.79	12	10.53	68	28.69
	3rd	78	22.22	33	28.95	45	18.99
	4th	45	12.82	12	10.53	33	13.92
	5th	34	9.69	11	9.65	23	9.70
	6th	11	3.13	2	1.75	9	3.80
Income	<8700	307	87.46	103	90.35	204	86.08
	≥ 8700	44	12.54	11	9.65	33	13.92
Residence	Rural	100	28.49	34	29.82	137	57.81
	Semi-urban	80	22.79	24	21.05	56	23.63
	Urban	100	28.49	56	49.12	44	18.57

Table 2
Behavioral and clinical characteristics of study participants in Ethiopia, 2024.

Variables	Category	Full sample		Good sleep quality		Poor sleep quality	
		N	%	N	%	N	%
Duration of smart phone used at bed	>2	237	67.52	88	77.19	149	62.87
	≤2	114	32.48	26	22.81	88	37.13
Year of digital devise use	<5	102	29.06	34	29.82	68	28.69
	5–10	171	48.72	58	50.88	113	47.68
	>10	78	22.22	22	19.30	56	23.63
Duration of daily use	≤2	124	35.33	33	28.95	91	38.40
	>2	227	64.67	81	71.05	146	61.60
Adjustment of screen display light at bedtime	No	178	50.71	55	48.25	123	51.90
	Yes	173	49.29	59	51.75	114	48.10
Position of phone during sleep	Near bed	185	52.71	78	68.42	107	45.15
	At bed	166	47.29	36	31.58	130	54.85
Purpose of use	Academic	153	14.53	99	28.95	54	7.59
	Communication	180	17.09	66	19.30	114	16.03
	Watching video	228	21.65	69	20.18	159	22.36
	Social media	338	41.60	43	27.19	295	48.52
	Gaming	144	5.13	15	4.39	39	5.49
Sleep duration in hours	≥8	160	45.58	61	53.51	99	41.77
	<8	191	54.42	53	46.49	138	58.23
Practicing physical exercise	No	237	67.52	57	50	180	75.95
	Yes	114	32.48	57	50	57	24.05
Caffeinated drinking	No	81	23.08	46	40.35	35	14.77
	Often	124	35.33	45	39.47	79	33.33
	Regular	146	41.60	23	20.18	123	51.90
Current tobacco smoker	No	316	90.03	108	94.74	208	87.76
	Yes	35	9.97	6	5.26	29	12.24
Current alcohol drinker	No	228	65	130	73.86	98	56
	Yes	123	35	46	26.14	77	44
History of disease	No	254	72.36	103	90.35	151	63.71
	Yes	97	27.64	11	9.65	86	36.29
Depression, Stress, anxiety disorder	No	250	71.23	102	89.47	148	62.45
	Yes	101	28.77	12	10.53	89	37.55
BMI (kg/m^2)	18.5–24.5	268	76.35	93	81.58	175	73.84
	<18.5	47	13.39	18	15.79	29	12.24
	24.5–30	24	6.84	2	1.75	22	9.28
	>30	12	3.42	1	0.88	11	4.64

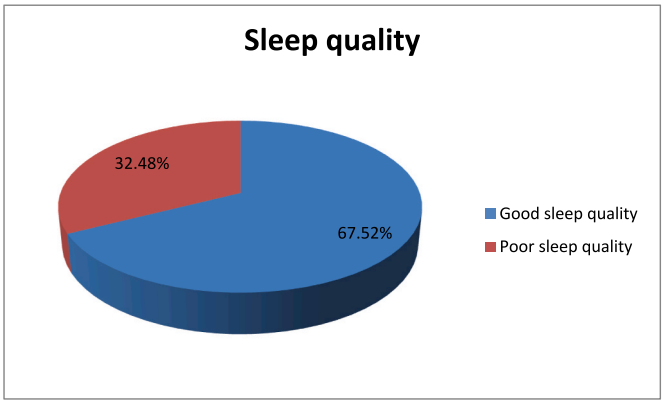


Fig. 1. prevalence of sleep quality among university medical students in Ethiopia, 2024.

consuming caffeinated beverages, current tobacco smoker, sleep duration, duration of smart phone used at bed, Position of phone during sleep, purpose of use, history of disease and phone position were considered as candidates for multivariable logistic regression at a significance level of p-value ≤0.2. However, in the subsequent multivariable logistic regression analysis, which was conducted at a 95 % confidence interval at a significance level of p-value ≤0.05 only sex, consuming caffeinated beverages, sleep duration, duration of smart phone used at bed, purpose of use, phone position and history of disease were identified as statistically significant variables. Being female increases the chance of developing poor sleep quality by 1.29 times (AOR = 2.91, 95 % CI; 1.54–5.51). Individuals who had drunk caffeinated substances had 4.68 times (95 % CI; 1.96–11.16) poor sleep quality than not drink at all but those drunk regularly had 8.96 AOR = 8.96, 95 % CI; 3.73–21.49) times more poor sleep quality. In addition, Participants who used smart phone at bed more than 2 h had AOR = 2.93 (95 % CI; 1.25–6.82) times high chance of developing poor sleep quality than the counter parts. Nonetheless, students who had used smartphone for social media had 2.93 AOR = 2.92(95 % CI; 1.17–7.22) times more likely to develop poor sleep quality. Moreover, participants who had put their phone at the bed during sleeping had 2.87 (95 % CI; 1.43–5.77) times more likely develop poor sleep quality than those who puts near their bed. Finally, individuals who reported history of disease had 2.53 AOR = 2.53 (95 % CI; 1.03–6.20) times high chance of developing poor sleep quality (Table 3).

3. Discussion

In this study, the overall prevalence of poor sleep quality among medical students was found to be 67.52 % (95 % CI: 62.42–72.24). This figure closely aligns with findings from comparable studies conducted in Egypt 64.2(7) and India 62.7 % [21]. The potential explanation for this similarity might be attributed to similarities in study design and the chosen cut-off point for assessing poor sleep quality. On the contrary, the prevalence of poor sleep quality in the current study was higher compared to studies conducted in India 48.87 % [22], Turkey 58.7 % [23], Vietnam (48.8 %) [24], Iran 61.7 % [25], china 9.8 % [26], Saudi Arabia [27] 41.7 %, Pakistan 61 % [28]. This higher proportion of poor sleep quality in the current study may be explained by the challenges faced by Ethiopian university students, such as low wages, high living costs, poor occupational health conditions, and unsatisfactory grievances and disputes, all of which contribute to stressful life conditions.

Table 3
The association between poor sleeps quality and smartphone utilization in multiple logistic regression analysis, Ethiopia, 2024.

Variable	Category	COR (95 % CI)	p- value	AOR (95 % CI)	p- value
Sex	Male	1	0.000	1	0.001
	Female	4.33(2.59–7.22)		2.91(1.54–5.51)	
Physical exercise	No	1	0.000	1	0.163
	Yes	0.32(0.19–0.51)		0.63(0.33–1.21)	
Caffeinated drinking	No	1	0.004	1	
	Often	2.31(1.30–4.09)	7.03	4.68(1.96–11.16)	0.001
	Regularly	7.03(3.76–13.14)		8.96(3.73–21.49)	0.000
Smoking	No	1	0.047	1	0.613
	Yes	2.51(1.01–6.23)		1.45(0.35–6.09)	
Duration of phone use at bed	≤2 h	1	0.000	1	0.013
	>2 h	1.75(1.08–4.59)		2.93(1.25–6.82)	
Purpose of use	Academic	1		1	
	Communication	3.17(1.45–6.89)	0.004	1.06(0.38–2.93)	0.910
	Watching video	4.22(1.98–8.98)	0.000	0.57(0.21–1.56)	0.272
	Social media	6.8(3.38–13.67)	0.000	2.92(1.17–7.22)	0.021
	Gaming	4.77(1.46–15.52)	0.010	2.74(0.59–12.52)	0.194
History of disease	No	1	0.000	1	0.043
	Yes	5.33(2.71–10.48)		2.53(1.03–6.20)	
Phone position	Near bed	1	0.000	1	0.003
	At bed	2.63(1.64–4.21)		2.87(1.43–5.77)	

Hosmer-lemeshow's goodness of fit test P-value was 0.78.

Existing evidence has highlighted that stressful life conditions may directly correlate with poorer sleep quality [29]. Furthermore, the ongoing financial distress in Ethiopia due to steadily rising inflation might also contribute to the high proportion of poor sleep quality, unlike studies conducted in Turkey, India, Vietnam, and Saudi Arabia, where participants live in relatively stable political and economic conditions [30,31]. Additionally, in the study conducted in China, they used a PSQI score of >7 in college students, while our study used a PSQI score of >5 (26). However, the prevalence of poor sleep quality was lower compared to studies conducted in Indonesia, which reported a rate of 78.85 % [32]. The possible reason for this difference could be that the study in Indonesia was conducted during the COVID-19 pandemic era. During this time, the incidence of smartphone addiction among students has increased due to reduced physical activity, leading to extended smartphone usage [33]. This increase in smartphone addiction has been associated with negative impacts such as symptoms of depression, anxiety disorders, and decreased sleep quality. Decreased sleep quality, in turn, can lead to insomnia, characterized by difficulties in maintaining both the quality and quantity of sleep [34,35].

Indeed, various factors can contribute to poor sleep quality among smartphone users of university students**.** Some of these factors include sex, duration of smart phone used at bed, purpose of use, phone position and history of disease. The odds of developing poor sleep quality were 1.29 times more likely in females than males. This was supported by studies conducted in Egypt [7], Malaysia [36]. This was largely due to hormonal fluctuations during the menstrual cycle. Estrogen and progesterone levels, which vary significantly throughout the cycle, can disrupt sleep. Elevated progesterone during the luteal phase, associated with symptoms like premenstrual syndrome (PMS) and premenstrual dysphoric disorder (PMDD), often leads to increased sleep latency and fragmented sleep. These hormonal changes, combined with menstrual discomfort and psychological symptoms, can exacerbate sleep disturbances [37].

Also, regular consumption of consuming caffeinated beverages has been identified as a significant contributing factor to sleep quality. Individuals who regularly consumed caffeinated drinks showed a 4.68 times higher likelihood of experiencing poor sleep quality compared to non-drinkers. This was supported by studies conducted in Egypt [7], Massachusetts [38]., Vietnam [20], and Saudi Arabia [4]. Caffeine impacts sleep primarily through two mechanisms: it blocks adenosine receptors, which are crucial for promoting sleep and relaxation, thus delaying the onset of sleepiness and extending the time it takes to fall asleep; and it stimulates cortisol production, a stress hormone that disrupts the natural sleep-wake cycle. Regular caffeine consumption also affects sleep parameters by increasing sleep latency (the time taken to fall asleep), reducing total sleep duration, and causing sleep fragmentation (frequent awakenings during the night). This fragmentation disrupts the continuity and restorative quality of sleep, leading to overall poorer sleep quality [38,39].

Additionally, individuals who spend more than 2 h using a smartphone in bed have a 2.93 times higher likelihood of experiencing poor sleep quality compared those used for less than 2 h. This finding is consistent with studies conducted in China [26], Saudi Arabia [27]. Several mechanisms have been proposed to elucidate the adverse impact of smartphone usage on sleep. Firstly, prolonged phone usage directly diminishes the amount of time allocated for sleep, particularly when smartphones are used before bedtime, resulting in inadequate sleep duration. Secondly, some users prefer browsing mobile websites prior to sleep, where inappropriate content may evoke feelings of tension and excitement, thereby impeding the initiation of sleep. Thirdly, excessive smartphone usage may disrupt sleep through physiological and psychological pathways [40]. For instance, exposure to the light emitted by smartphone screens and radiation from the device during bedtime could disrupt the onset time and secretion of melatonin, consequently leading to disturbances in the sleep-wake rhythm [41,42].

Nonetheless, Social media was the most widely used social service among study participants. Students who had used smartphone for social media had 2.93 times more likely to develop poor sleep quality than those used for academic purpose. it was supported by studies conducted in Saudi Arabia [27], Philippines [15], Egypt [7]. Engaging with social media before bedtime can delay the onset of sleep and reduce the total time spent sleeping. Moreover, the content on social media platforms often stimulates mental activation, making it difficult to fall asleep [43]. Finally, individuals who reported history of disease had 2.53 times high chance of developing poor sleep quality. Diseases can affect sleep quality through various mechanisms [44]. Symptoms like pain, discomfort, or respiratory issues can disrupt sleep, making it difficult to fall or stay asleep. Medications used to treat diseases may also have side effects such as drowsiness, insomnia, or restless legs, which can interfere with sleep [45]. Additionally, physical limitations caused by diseases, such as mobility issues or breathing problems can make it uncomfortable to find a restful sleeping position. Furthermore, some diseases disrupt the body's natural sleep-wake cycle, leading to disturbances in sleep patterns [46].

4. Limitation of the study

The limitations of this study are as follows: First, due to the cross-sectional design, it was not possible to examine causal associations between smartphone use before bed, poor sleep quality, and academic disruption. Second, sleep quality and duration of smartphone use were self-reported by participants, which may have led to recording bias.

5. Conclusion and recommendation

This study revealed that smartphone use at bedtime was highly prevalent among the surveyed medical students. Additionally, poor sleep quality was prevalent among study participants, and it was significantly higher among bedtime smartphone users, highlighting this as a significant health issue that negatively influences academic activities. Therefore, it is crucial to educate medical students about the negative effects of smartphone use before bedtime on sleep and the importance of adequate sleep for good academic performance.

CRediT authorship contribution statement

Dereje Esubalew: Writing – original draft, Software, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Amare Mebrat Delie:** Writing – review & editing, Visualization, Validation, Supervision, Investigation, Data curation. **Liknaw Workie Limenh:** Writing – review & editing, Visualization, Supervision, Methodology, Funding acquisition, Conceptualization. **Nigus Kassie Worku:** Writing – review & editing, Visualization, Validation, Software, Investigation, Data curation. **Eneyew Talie Fenta:** Writing – review & editing, Visualization, Supervision, Software, Resources, Funding acquisition, Data curation. **Mickiale Hailu:** Writing – review & editing, Visualization, Validation, Supervision, Resources, Data curation. **Alemwork Abie:** Writing – review & editing, Visualization, Validation, Supervision, Software, Data curation, Conceptualization. **Molla Getie Mehari:** Visualization, Validation, Supervision, Software, Methodology, Data curation. **Tenagnework Eseyneh Dagnaw:** Writing – review & editing, Visualization, Validation, Supervision, Software, Resources, Data curation. **Mihret Melese:** Writing – review & editing, Visualization, Validation, Supervision, Software, Methodology, Funding acquisition, Conceptualization.

Informed consent

Informed consent was obtained from all participants.

Data availability

The datasets utilized and/or analyzed in this study can be obtained from the corresponding author upon a reasonable request.

Ethical approval

This research received approval from the Institutional Review Board (IRB) of Ambo University, Ethiopia (IRB/1405/2024). Written consent obtained from all participants and the study's objectives and nature thoroughly communicated and explained to the respondents before providing consent. The principles of the Declaration of Helsinki strictly adhered to throughout the study. Participants' names not collected, but addresses and other relevant information obtained with explicit permission. Subsequently, the authors upheld confidentiality throughout the study period and securely stored this information. Once the study's objectives are achieved, the data will be securely disposed of to safeguard participants' anonymity and privacy. All the data analyses were conducted anonymously.

Consent for publication

Not applicable.

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Not applicable.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.sleepx.2024.100134.

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